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Experiment-9

QUESTION 1

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds.

1. Problem Understanding

The program takes the total time in **seconds** from the user and converts it into **days, hours, minutes, and seconds**.

Input: Time in seconds

Output: Days, hours, minutes and seconds

Conversion used:

1 day = 86400 seconds

1 hour = 3600 seconds

1 minute = 60 seconds

Example:

Input: 90061 seconds

Output: 1 day, 1 hour, 1 minute, 1 second

2. Algorithm

1. Start.
2. Input the total number of seconds.
3. Find days by dividing total seconds by 86400.
4. Find the remaining seconds after calculating days.
5. Find hours by dividing the remaining seconds by 3600.
6. Find the remaining seconds after calculating hours.
7. Find minutes by dividing the remaining seconds by 60.

8. The remaining value gives the seconds.
9. Display days, hours, minutes and seconds.
10. Stop.

3. Pseudocode

```

START
INPUT total_seconds
days = total_seconds / 86400
remaining = total_seconds % 86400
hours = remaining / 3600
remaining = remaining % 3600
minutes = remaining / 60
seconds = remaining % 60
PRINT days, hours, minutes, seconds
STOP

```

4. Flowchart

```

START

Enter total seconds

days = total_seconds / 86400

remaining = total_seconds % 86400

hours = remaining / 3600

remaining = remaining % 3600

minutes = remaining / 60

seconds = remaining % 60

Display days, hours, minutes and seconds

STOP

```

5. Test Case Table

Test Case	Input	Expected Output	Actual Output	Results
1	90061	1 day, 1 hour, 1 minute, 1 second	1 day, 1 hour, 1 minute, 1 second	Pass
2	3600	0 days, 1 hour, 0 minutes, 0 seconds	0 days, 1 hour, 0 minutes, 0 seconds	Pass
3	60	0 days, 0 hours, 1 minute, 0 seconds	0 days, 0 hours, 1 minute, 0 seconds	Pass

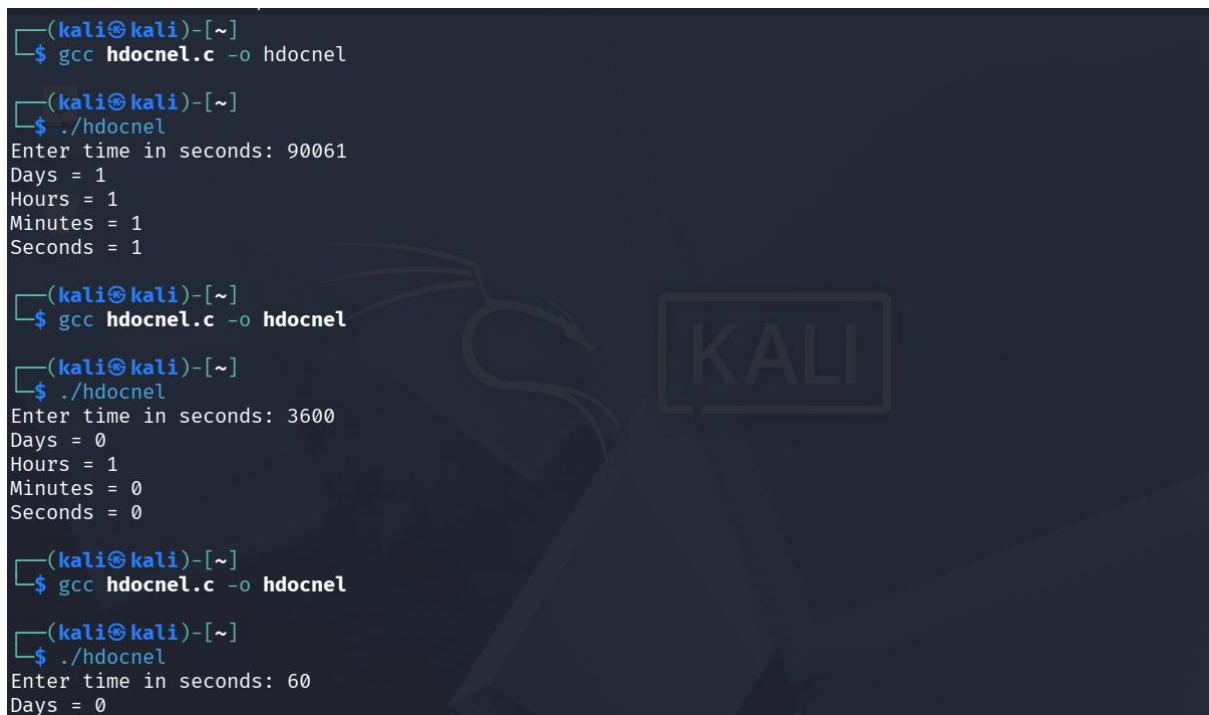
4	86400	1 day, 0 hours, 0 minutes, 0 seconds	1 day, 0 hours, 0 minutes, 0 seconds	Pass
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6. C Code

```
#include <stdio.h>
int main()
{
    int total_seconds, days, hours, minutes, seconds, remaining;
    printf("Enter time in seconds: ");
    scanf("%d", &total_seconds);
    days = total_seconds / 86400;
    remaining = total_seconds % 86400;

    hours = remaining / 3600;
    remaining = remaining % 3600;
    minutes = remaining / 60;
    seconds = remaining % 60;
    printf("Days = %d\n", days);
    printf("Hours = %d\n", hours);
    printf("Minutes = %d\n", minutes);
    printf("Seconds = %d\n", seconds);
    return 0;
}
```

7. Compilation and Execution



```
(kali㉿kali)-[~]
└─$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
└─$ ./hdocnel
Enter time in seconds: 90061
Days = 1
Hours = 1
Minutes = 1
Seconds = 1

(kali㉿kali)-[~]
└─$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
└─$ ./hdocnel
Enter time in seconds: 3600
Days = 0
Hours = 1
Minutes = 0
Seconds = 0

(kali㉿kali)-[~]
└─$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
└─$ ./hdocnel
Enter time in seconds: 60
Days = 0
```

```
Hours = 0
Minutes = 1
Seconds = 0

(kali㉿kali)~[~]
$ gcc hdocnel.c -o hdocnel

(kali㉿kali)~[~]
$ ./hdocnel
Enter time in seconds: 86400
Days = 1
Hours = 0
Minutes = 0
Seconds = 0
```

QUESTION 2

Write a program to input a year and check whether it is a leap year or not.

1. Problem Understanding

The program takes a **year** from the user and checks whether the given year is a **leap year or not**.

A year is a leap year when:

- It is divisible by **400**, OR
- It is divisible by **4 but not by 100**.

For example:

2024 → Leap Year
2025 → Not a Leap Year
1900 → Not a Leap Year
2000 → Leap Year

2. Algorithm

1. Start.
2. Input the year.
3. Check if the year is divisible by 400.
4. If yes, display "Leap Year".
5. Otherwise, check if the year is divisible by 4 and not divisible by 100.
6. If yes, display "Leap Year".
7. Otherwise, display "Not a Leap Year".
8. Stop.

3. Pseudocode

```
START
INPUT year
IF year % 400 == 0
```

```
    PRINT "Leap Year"
ELSE IF year % 4 == 0 AND year % 100 != 0
    PRINT "Leap Year"
ELSE
    PRINT "Not a Leap Year"

STOP
```

4. Flowchart

```
START
Input year
Is year divisible by 400?
Yes

Display "Leap Year"

STOP
No
Is year divisible by 4 AND not divisible by 100?

Yes

Display "Leap Year"

STOP
No

Display "Not a Leap Year"

STOP
```

5. Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	2024	Leap Year	Leap Year	Pass
2	2025	Not a Leap Year	Not a Leap Year	Pass
3	2000	Leap Year	Leap Year	Pass
4	1900	Not a Leap Year	Not a Leap Year	Pass
5	2028	Leap Year	Leap Year	Pass

6. C Code

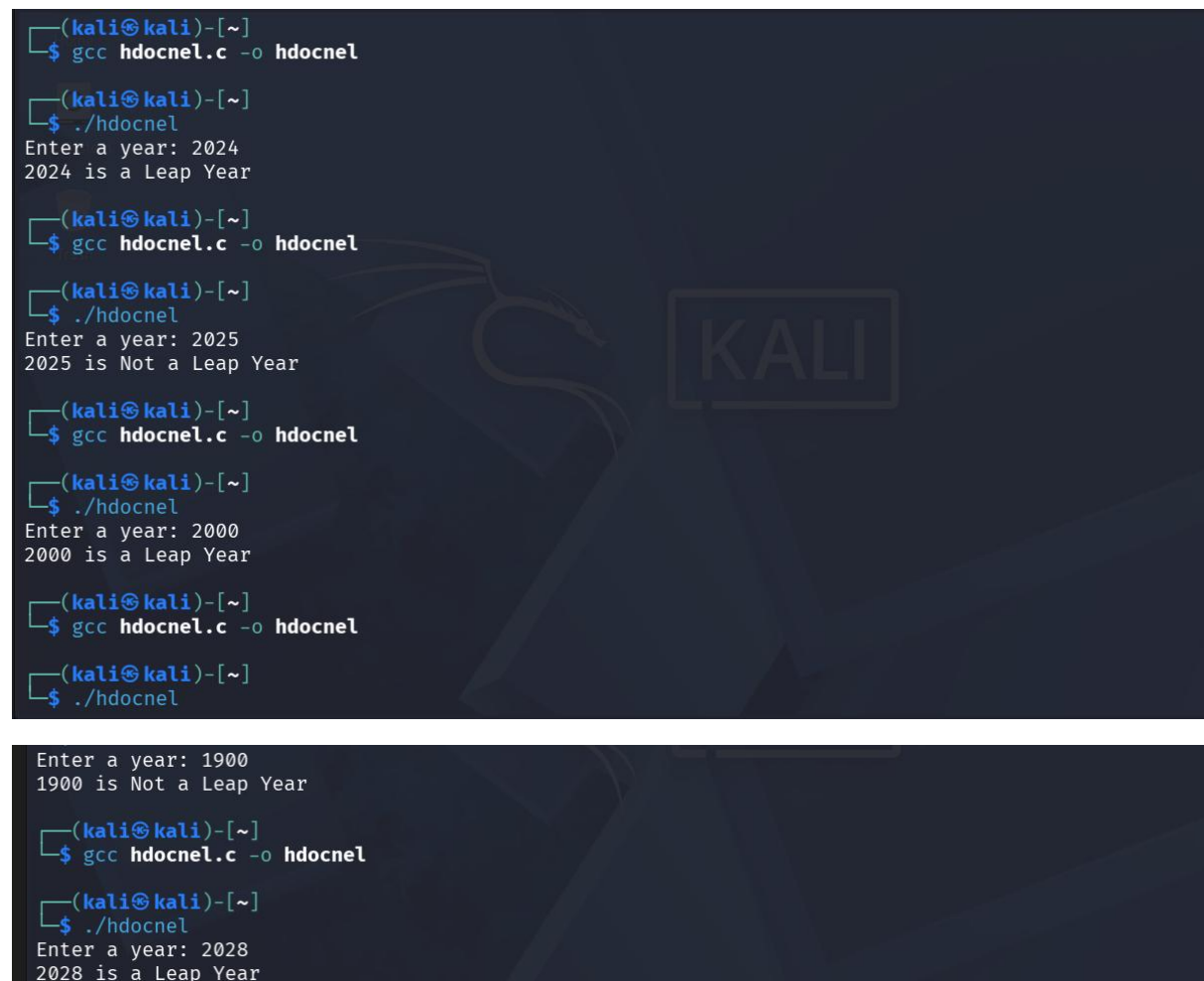
```

#include <stdio.h>
int main()
{
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if (year % 400 == 0)
    {
        printf("%d is a Leap Year\n", year);
    }
    else if (year % 4 == 0 && year % 100 != 0)
    {
        printf("%d is a Leap Year\n", year);
    }
    else
    {
        printf("%d is Not a Leap Year\n", year);
    }

    return 0;
}

```

7. Compilation and Execution



```

(kali㉿kali)-[~]
$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
$ ./hdocnel
Enter a year: 2024
2024 is a Leap Year

(kali㉿kali)-[~]
$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
$ ./hdocnel
Enter a year: 2025
2025 is Not a Leap Year

(kali㉿kali)-[~]
$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
$ ./hdocnel
Enter a year: 2000
2000 is a Leap Year

(kali㉿kali)-[~]
$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
$ ./hdocnel
Enter a year: 1900
1900 is Not a Leap Year

(kali㉿kali)-[~]
$ gcc hdocnel.c -o hdocnel

(kali㉿kali)-[~]
$ ./hdocnel
Enter a year: 2028
2028 is a Leap Year

```